

# Interacting Spin-Chain Reproduction Report for the Nonsymmorphic DTC Model

Local CUDA validation, H100 long-time run, and next-run guidance

June 4, 2026

## Abstract

This report documents the interacting-stage reproduction of the nonsymmorphic discrete-time-crystal model. The calculation adds nearest-neighbor Ising interactions to the previously reproduced two-level drive, validates the implementation on the local RTX 4090 GPU, and incorporates an  $L = 8$  long-time H100 run for the stroboscopic diagnostic  $Z(n)$ .

## 1 Scope

The goal is to reproduce the interacting behavior discussed around Fig. 4(b–d) of Hu, Fu, Li, and Shen, *Physical Review Research* **5**, L032024 (2023). The local calculation is intentionally small: it verifies the model, observable definitions, parameter choices, and qualitative DTC response. The cluster calculation then focuses on the  $L = 8$  long-time stroboscopic behavior.

## 2 Hamiltonian

The interacting chain Hamiltonian is

$$H(t) = \sum_{i=1}^L H_{0,i}(t) + J \sum_{i=1}^{L-1} \sigma_x^i \sigma_x^{i+1}, \quad (1)$$

with open boundary conditions and

$$H_{0,i}(t) = \frac{1}{2} \Omega \sin(t) \sigma_x^i + \Omega \sin^2\left(\frac{t}{2}\right) \sigma_y^i. \quad (2)$$

The dimensionless convention is

$$\hbar = 1, \quad \omega = 1, \quad T = 2\pi, \quad \Omega = \alpha/8.$$

The interaction direction is  $\mu = x$ , matching the main-text interacting figure. The interaction strength is

$$J/\hbar\omega = 0.2.$$

Two values of  $\alpha$  are used:

$$\alpha = 81.60, \quad \alpha_{13} = 80.04624211689759.$$

The first is the detuned value used in the main-text finite-size lifetime plot, while the second is the refined special value near the paper's tabulated  $\alpha_{13} \approx 80.07$ .

### 3 Initial State and Observables

The initial state is the product state polarized in the  $+x$  direction,

$$|\psi(0)\rangle = \bigotimes_{i=1}^L |+\rangle_i. \quad (3)$$

The primary observable is the average  $x$ -magnetization,

$$m_x(t) = \frac{1}{L} \sum_{i=1}^L \langle \psi(t) | \sigma_x^i | \psi(t) \rangle. \quad (4)$$

At stroboscopic times  $t = nT$ , the reported DTC order diagnostic is

$$Z(n) = (-1)^n m_x(nT). \quad (5)$$

If the response is period-doubled and stable,  $Z(n)$  remains positive and slowly varying over a long time window.

### 4 Numerical Method

The local calculation uses an exact state vector of dimension  $2^L$ . Instead of constructing dense Hamiltonian matrices, the script applies the Hamiltonian to the state by precomputed bit-flip index arrays:

- $\sigma_x^i$  flips one bit;
- $\sigma_y^i$  flips one bit with the appropriate  $\pm i$  phase;
- $\sigma_x^i \sigma_x^{i+1}$  flips two neighboring bits.

This keeps the implementation transparent while allowing small GPU simulations. Time evolution uses fourth-order Runge–Kutta with per-step normalization. The normalization controls accumulated RK4 norm drift during many-period runs; the saved diagnostics still report the observed norm error after normalization.

For the long-time HPC run, direct step-by-step evolution to  $10^5$ – $10^8$  periods would be inefficient and would produce unnecessarily large output files. Instead, the code constructs the one-period Floquet operator for  $L = 8$  using midpoint matrix exponentials over one drive period, diagonalizes this  $256 \times 256$  unitary matrix, and evaluates  $Z(n)$  at logarithmically spaced stroboscopic periods. This is why the  $L = 8$  cluster job can finish quickly while still probing the relevant period range.

### 5 Local CUDA Validation

The validation run used the local NVIDIA GeForce RTX 4090. The time trace used  $L = 10$ , 40 periods, and 80 time steps per period. The stroboscopic scan used  $L = 4, 6, 8, 10$ , 80 periods, and 40 time steps per period.

### 6 Results

Figure 1 shows the first 40 periods for  $L = 10$ . The two parameter values overlap closely in this local time window. Most importantly, the Fourier spectrum has a dominant peak at  $\tilde{\omega}/\omega \simeq 0.5$ , matching the paper’s statement that interactions stabilize a cleaner subharmonic response.

Table 1: Local interacting validation diagnostics.

| Quantity                         | $\alpha = 81.60$      | refined $\alpha_{13}$ |
|----------------------------------|-----------------------|-----------------------|
| Trace length                     | 10                    | 10                    |
| Dominant $\tilde{\omega}/\omega$ | 0.4998                | 0.4998                |
| Maximum norm error               | $1.19 \times 10^{-7}$ | $1.19 \times 10^{-7}$ |
| Minimum $m_x(t)$ over trace      | -0.9932               | -0.9932               |
| Maximum $m_x(t)$ over trace      | 1.0000                | 1.0000                |

Figures 2 and 3 show the stroboscopic diagnostic  $Z(n)$ . The smallest chain,  $L = 4$ , has strong finite-size oscillations and decays within the local window. The larger local chains, especially  $L = 6, 8, 10$ , retain  $Z(n) \approx 0.95$ – $0.98$  across the 80 simulated periods. This verifies that the local implementation captures the interaction-stabilized period-doubled response, while longer-time finite-size scaling should be run on the cluster.

## 7 HPC Long-Time Result for $L = 8$

The cluster run used one NVIDIA H100 80GB GPU on node `gpuh14`. The Slurm launcher excluded `gpuh01`. The time-trace part used  $L = 8$ , 40 periods, and 120 time steps per period. The stroboscopic part used the Floquet logarithmic-sampling mode:

$$\alpha = 81.60 : n_{\max} = 10^5, \quad \alpha_{13} = 80.04624211689759 : n_{\max} = 10^8.$$

Both cases retain a dominant Fourier peak at  $\tilde{\omega}/\omega = 0.4998958550$  in the first 40 periods, consistent with period doubling.

Table 2: HPC  $L = 8$  stroboscopic diagnostics. The threshold periods are the first sampled  $n$  where  $Z(n)$  drops below the listed value.

| Quantity               | $\alpha = 81.60$      | refined $\alpha_{13}$ |
|------------------------|-----------------------|-----------------------|
| Maximum sampled period | $10^5$                | $10^8$                |
| Number of samples      | 1793                  | 1801                  |
| Maximum norm error     | $2.39 \times 10^{-5}$ | $1.64 \times 10^{-5}$ |
| First $Z(n) < 0.8$     | 12                    | 82839                 |
| First $Z(n) < 0.5$     | 2018                  | 138896                |
| First $Z(n) < 0$       | 3642                  | 212790                |
| Final sampled $Z(n)$   | 0.6633                | -0.0351               |

Figure 4 shows the long-time  $L = 8$  stroboscopic result on a logarithmic period axis. The detuned case,  $\alpha = 81.60$ , loses the high- $Z$  plateau quickly and crosses below  $Z = 0.8$  at  $n = 12$  in the sampled data. The refined value  $\alpha_{13}$  remains near  $Z \simeq 1$  through much longer times and first drops below  $Z = 0.8$  near  $n = 8.3 \times 10^4$ . This confirms the expected qualitative trend from Fig. 4: tuning to the special  $\alpha_n$  strongly enhances the lifetime, even though the finite  $L = 8$  system eventually shows revivals and sign-changing oscillations.

## 8 Generated Files

The local stage generated:

- `reproduce_interacting.py`;

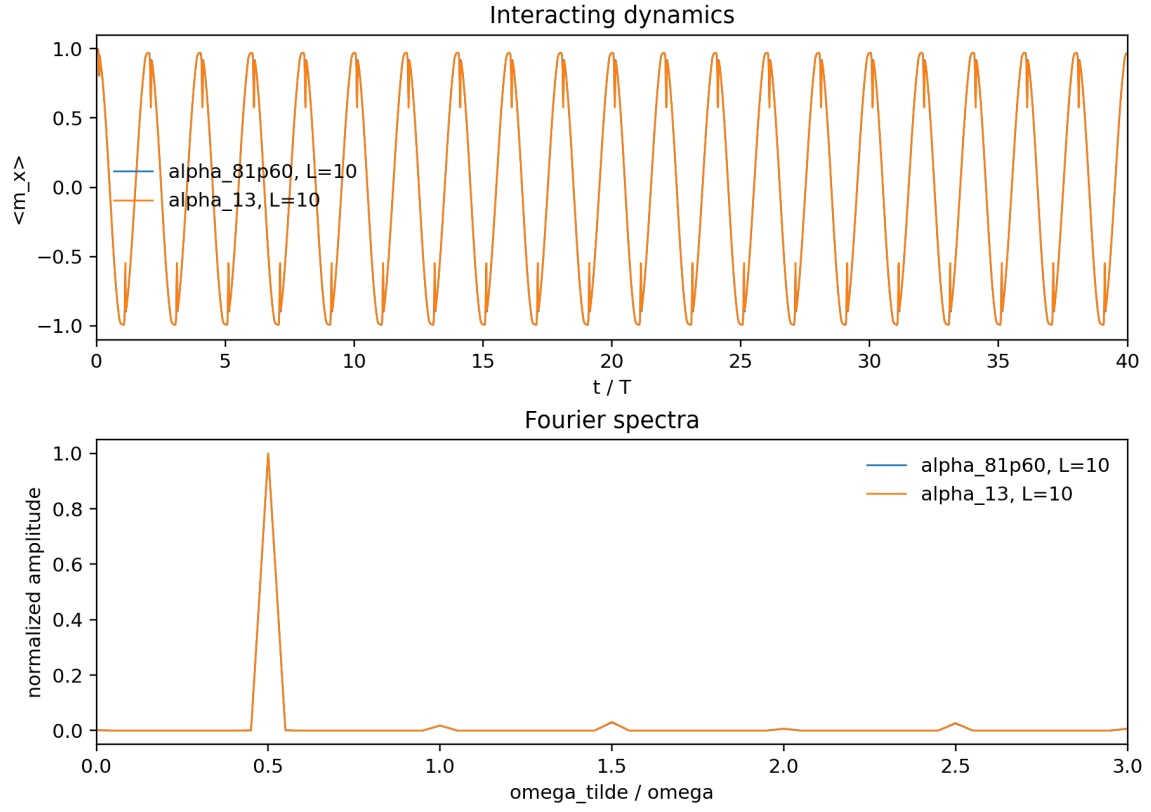


Figure 1: Interacting  $m_x(t)$  and Fourier spectra for  $L = 10$ ,  $J = 0.2$ , and the two  $\alpha$  values.

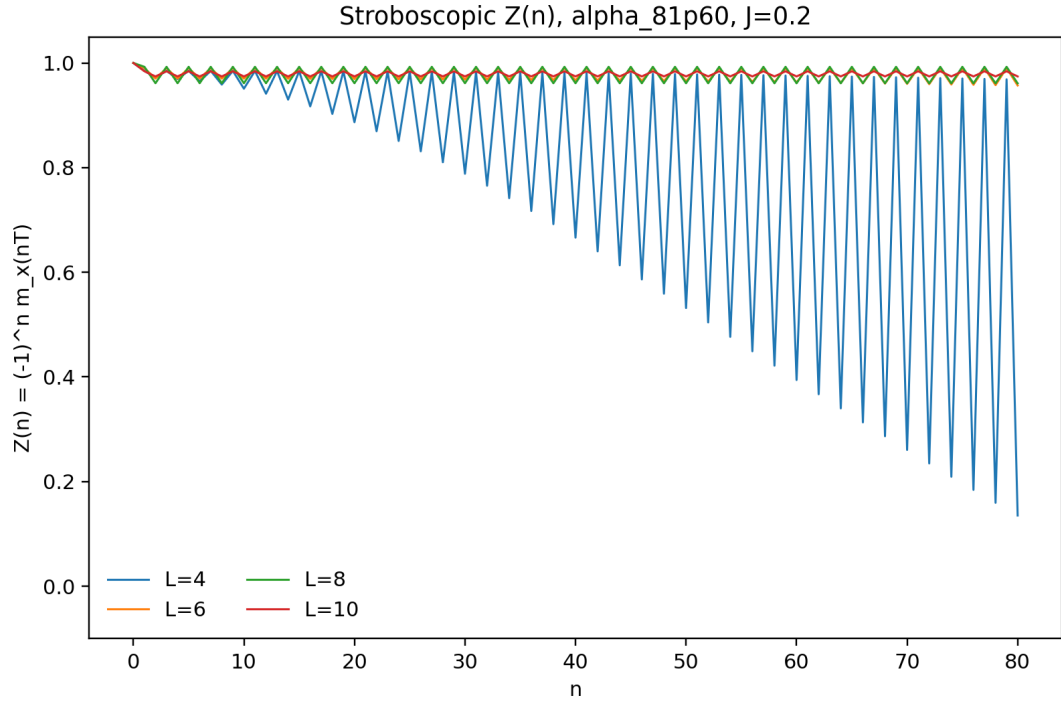


Figure 2: Stroboscopic  $Z(n)$  for  $\alpha = 81.60$ ,  $J = 0.2$ , and small chain lengths.

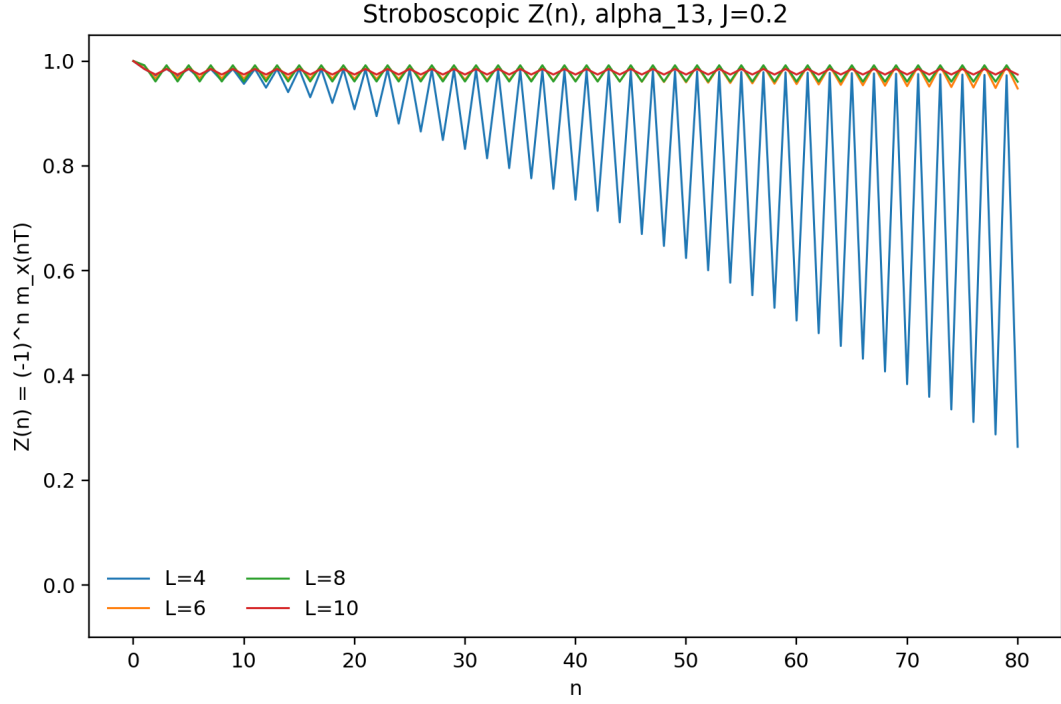


Figure 3: Stroboscopic  $Z(n)$  for refined  $\alpha_{13}$ ,  $J = 0.2$ , and small chain lengths.

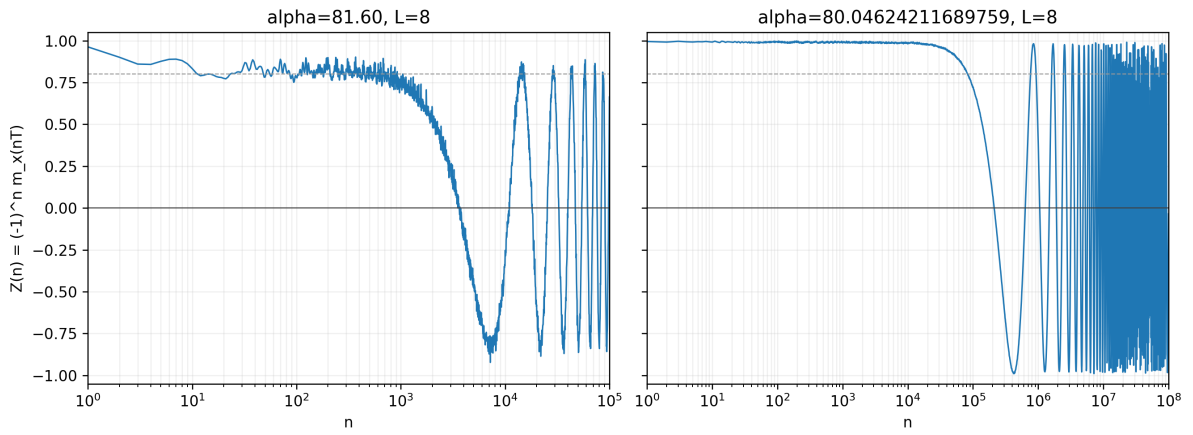


Figure 4: HPC long-time stroboscopic  $Z(n)$  for  $L = 8$ ,  $J = 0.2$ . The horizontal dashed line marks  $Z = 0.8$ , a useful operational lifetime threshold.

- `params/interacting_params_local.json`;
- `params/interacting_params_hpc.json`;
- `data/mx_trace_*.csv`;
- `data/mx_spectrum_*.csv`;
- `data/z_stroboscopic_*.csv`;
- `data/diagnostics.json`;
- `figures/mx_trace_comparison_l10.png`;
- `figures/z_stroboscopic_alpha_81p60.png`;
- `figures/z_stroboscopic_alpha_13.png`;
- `figures/hpc_z_l8_longtime.png`;
- `run_interacting_hpc.sh`.

## 9 HPC Package and Larger-System Parameters

The upload package contains the Python script, local and HPC parameter files, a bash launcher, and empty `logs/` and `results/` directories. The HPC launcher includes

```
#SBATCH --exclude=gpuh01
```

so that the old-CUDA `gpuh01` node is excluded.

### How to Change the Next Run

The main file to edit is `params/interacting_params_hpc.json`. For a larger system, change only the stroboscopic length list and, if needed, the maximum sampled periods:

```
"stroboscopic": {
  "lengths": [10],
  "periods_by_alpha": {
    "alpha_81p60": 100000,
    "alpha_13": 10000000000
  },
  "steps_per_period": 120,
  "sampling_mode": "log_floquet",
  "linear_prefix_periods": 200,
  "log_sample_count": 1600
}
```

For  $L = 10$ , the Floquet matrix dimension is  $2^{10} = 1024$ , which is still realistic on an H100 but more expensive than the  $L = 8$  run. For  $L = 12$ , the matrix dimension is 4096, and the dense Floquet path may become memory- and time-intensive because it repeatedly exponentiates dense matrices. A pragmatic next step is therefore  $L = 10$  before attempting  $L = 12$ .

## 10 Limitations and Next Step

The current long-time run is only for  $L = 8$ , so it does not yet reproduce the full finite-size lifetime scaling inset of Fig. 4(c,d). The next step is to repeat the same Floquet logarithmic-sampling workflow for  $L = 10$ , then compare the threshold period  $n_c$  against the  $L = 8$  result. If the  $L = 10$  dense Floquet run remains affordable,  $L = 12$  can be attempted afterward.